

Sneha Sebastian

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PROFILE

MSc graduate in Data Science with a strong academic foundation in mathematics, statistics, and machine learning, aspiring to pursue a career in higher education and research. Experienced in teaching support, academic projects, and applied data analysis using large-scale real-world datasets. Actively involved in interdisciplinary research at the intersection of data science and healthcare, with hands-on experience in statistical modeling, machine learning, and model interpretability. Proficient in Python, R, SQL, and data visualization tools, with a strong commitment to student mentoring, academic excellence, and outcome-based education.

Skills

- **Subject Expertise:** Probability and Statistics, Linear Algebra, Machine Learning, Data Analytics, Research Methodology
- **Programming & Statistical Tools:** Python, R, SQL, STATA (basic)
- **Machine Learning & Data Analysis:** Regression Models, Classification Algorithms, Feature Selection, Model Evaluation, Data Preprocessing
- **Data Visualization & Reporting:** Tableau, Power BI, Matplotlib, Seaborn
- **Research Skills:** Literature Review, Survey Data Analysis, Statistical Interpretation, Technical and Academic Writing
- **Teaching & Academic Skills:** Classroom Instruction Support, Lab Demonstrations, Student Mentoring, Assignment Evaluation, Academic Presentations
- **Soft Skills:** Effective Communication, Classroom Engagement, Leadership, Team Collaboration, Time Management

Education

M.Sc in Data Science and Analytics [First Rank] July 2023 - June 2025
Mahatma Gandhi University Kottayam, India

- **CGPA:** 9.65
- **Coursework:** Data Visualization, Data Mining, Artificial Intelligence, Machine learning, Natural Language Processing, Time series Analysis

B.Sc in Mathematics Jun 2019 - Mar 2022
Government College Kasaragod, Kannur University, India

- **Percentage:** 93.16%
- **Coursework:** Probability and Statistics, Linear Algebra, Complex Analysis, Graph Theory, Real Analysis

Experiences

International Students Project Programme (ISPP) 2025 [Remote] Feb 2025– Aug 2025

- Part of an international research program conducted by Phuket Rajabhat University, Thailand.
- Collaborating with an international, multidisciplinary team to develop computational and integrative frameworks for epidemiological modelling using statistical and machine learning methods.

Research Intern at Jawaharlal Nehru University, New Delhi Dec 2024 – May 2025

- Undertaken as part of final semester MSc project.
- Gained experience in machine learning-based feature selection, imputation techniques, and statistical modeling to analyze health data and evaluate model performance.

IBM SkillsBuild Winter Internship Program [Remote]

Dec 2024 – Jan 2025

- Successfully completed training on “Turning Data into Decisions using Python” with CSRBOX, covering data preprocessing, analysis, and visualization techniques.
- Gained hands-on experience in Python for data analytics, covering statistical analysis, decision-making, and real-world applications.

IBM SkillsBuild Summer Internship Program [Remote]

July 2024 – Aug 2024

- IBM SkillsBuild Summer Internship Program with CSRBOX on Artificial Intelligence.
- Gained hands-on experience in AI technologies like NLP and machine learning, contributing to the development of an AI-powered chatbot.

Projects

Machine Learning-Based Identification of Fertility Determinants in Sikkim: A Feature Selection and Imputation-Driven Approach

- A data-driven project aimed at identifying key factors influencing declining fertility in Sikkim using machine learning techniques, with a focus on feature selection and imputation strategies to enhance model accuracy.
- Technologies Used: Python, Google Colab, SPSS

Machine Learning-Based Glioma Classification and Subtype Discovery Using Hybrid LARF and Stratified Clustering on Clinical and Genomic Data

- A machine learning project focused on classifying glioma grades using clinical and genomic data, enhancing accuracy through LARF-based feature selection and model interpretability techniques.
- Technologies Used: Python, Google Colab

Analysing Flight Transactions using Tableau

- Using Agoda’s flight transaction data to provide comprehensive insights into flight sales and operations.
- Technologies Used: Tableau

Crop Recommendation System using Machine Learning

- Developed a Machine Learning-based system to recommend crops by analyzing soil factors, weather conditions, and rainfall, enabling optimized crop selection for higher yields.
- Technologies Used: Python, HTML, Visual Studio Code, Google Colab

Certificates

- **Workshops:** PowerBI (2025), Deep Learning & LLMs (2024), Web Scraping with Python (2024), Data Analysis using R (2023), Bayesian Computing & MCMC (2023)
- **Online Courses:** NLP for Speech and Text (LinkedIn Learning, 2025), ML & AI Foundations (LinkedIn Learning, 2025)